

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459191.64148 +/- 0.00022 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.001 +/- 0.033
Transit depth (Rp/Rs)^2	6.1e-05 +/- 0.0052 [%]
Orbital Inclination (inc)	89.34 +/- 1.45
Airmass coefficient 1 (a1)	1.23 +/- 0.14
Airmass coefficient 2 (a2)	-0.114 +/- 0.043
Scatter in the residuals of the lightcurve fit is	11.54 %
Best Comparison Star	#3 - [302, 409]
Optimal Aperture	6.27
Optimal Annulus	7.11
Transit Duration (day)	0.0815 +/- 0.002

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.001 ± 0.033 vs 0.1301 (deviation 1.93σ)
Duration fit vs published	0.0815 d vs 0.0908 d (deviation 2.3σ)
Mid-time O–C	0.3 min
Depth SNR	0.0
Residual scatter	11.54 %

Light curve

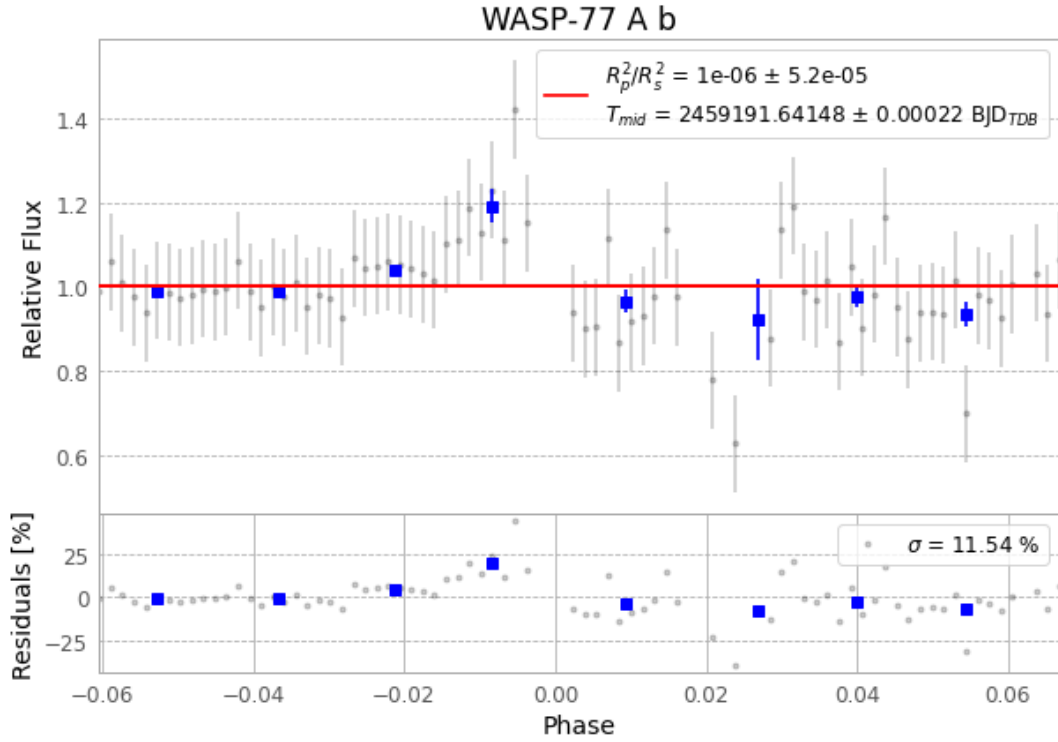


Figure 1 — FinalLightCurve_WASP-77 A b_2020-12-07.png (SHA-256 5823be6ccbe41c2a...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [620, 149], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 bdbcc633e2a5b9bd...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

84 FITS frames (55 MB), WASP-77201208011812.FITS → WASP-77201208052712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-77-201208.log (c960a8d28cc1...), FinalLightCurve_WASP-77 A b_2020-12-07.png (5823be6ccbe4...), FinalLightCurve_WASP-77 A b_2020-12-07.csv (68e111860ea0...)

Compute resources

Wall time 175.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 04:49Z.